

Diversity-Driven Ensemble Weighting for Probabilistic Price Forecasting: Introducing the Synth Variance-Based Weighted Metamodel

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Abstract

The SYNTH Subnet is a decentralised forecasting network in which miners compete by submitting probabilistic price path ensembles for cryptocurrency and equity assets. We introduce the **Synth Weighted Metamodel**, a data-driven ensemble scheme in which miner weights are optimised by minimising the Quasi-Likelihood (QLIKE) loss on hourly realized variance forecasts. Backtests spanning late January to mid-March 2026 show that the new model reduces out-of-sample forecast error by 1.2% in the 24-hour competition and 13.1% in the HFT competition, achieves superior distributional calibration across all evaluation horizons, and produces better-calibrated prediction-market probabilities. Weights are concentrated on miners from a diverse set of independent operators, providing richer forecasting signal for trading applications.

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1 Introduction

1.1 The Synth Subnet

The SYNTH Subnet (Mode Network / Synth Data, 2024; Synth Data, 2025) is a decentralised forecasting network built on the Bittensor protocol (Bittensor, 2023). Rather than asking participants to submit single-point predictions, SYNTH focuses on the full *distributional* forecast: miners are required to generate $N_{\text{sim}} = 1,000$ simulated price paths for each queried asset, covering a specified future horizon at a fixed time increment.

Two parallel competitions are currently active:

1. **24-Hour (low-frequency) competition.** Miners submit 1,000 price paths at 5-minute increments over a 24-hour horizon for a set of cryptocurrency and tokenised equity assets (BTC, ETH, XAU, SOL, SPYX, NVDA, AAPL, TSLA, GOOGL). Prompts are issued roughly every hour. This competition accounts for 50% of total network emissions.
2. **1-Hour HFT (high-frequency) competition.** Miners submit paths at 1-minute increments over a 1-hour horizon for BTC, ETH, XAU, and SOL. Also responsible for 50% of total emissions, this track has been designed to support short-horizon trading applications requiring rapid model updates.

Miners are identified by a unique integer *UID* and communicate with validators through their network address (*axon*). The validator evaluates each submission using the *Continuous Ranked Probability Score* (CRPS; Gneiting and Raftery, 2007; Matheson and Winkler, 1976), a proper scoring rule that rewards both calibration and sharpness. For an observed value x and an ensemble of N simulated paths $y_1, \dots, y_N$, the CRPS is:

$$\text{CRPS}(F_N, x) = \frac{1}{N} \sum_{n=1}^N |y_n - x| - \frac{1}{2N^2} \sum_{n=1}^N \sum_{m=1}^N |y_n - y_m|, \quad (1)$$

where the first term measures ensemble-observation discrepancy and the second penalises excess ensemble spread. CRPS values are computed on *price changes in basis points* across multiple sub-horizons (5 min, 30 min, 3 h, and 24 h for the low-frequency track), and summed to form the per-prompt score.

A rolling 10-day weighted average of per-prompt scores constitutes the *leaderboard score* $L_i(t)$ for miner i :

$$L_i(t) = \frac{\sum_{j: t-t_j \leq 10 \text{ d}} S_{i,j} w_{k,j}}{\sum_{j: t-t_j \leq 10 \text{ d}} w_{k,j}}, \quad (2)$$

where $S_{i,j}$ is miner i 's score at prompt j and $w_{k,j}$ is the asset weight for asset k at prompt j . Emission allocations are then determined by a softmax over leaderboard scores.

1.2 The Original Metamodel

SYNTH’s products—including publicly released volatility forecasts and prediction-market probability estimates—have been powered by an in-house *metamodel*: a procedure that aggregates individual miner paths into a single, representative forecast distribution. The original approach, which we refer to as the **Synth Classic Metamodel** throughout this paper, operates as follows:

1. Select the top- K miners by cumulative leaderboard reward over a recent look-back window (typically the last several days).
2. Pool all selected miners’ paths uniformly, forming a combined empirical distribution.

This scheme is computationally lightweight, transparent, and has proven effective in practice. However, it is built on a purely reward-based heuristic rather than a principled statistical criterion. In particular, it does not explicitly account for the *diversity* of the constituent miners’ modelling approaches—a critical consideration when a subset of miners may be highly correlated clones operated by the same entity.

1.3 Motivation for a New Approach

Two empirical observations, presented in full in Section 2, motivated us to move beyond the heuristic baseline.

First, a clustering analysis of miner behaviour over the past two months reveals that the SYNTH network is effectively controlled by a small number of operators, each running multiple miners whose predictions are nearly identical. The largest identified group alone comprises 13 axons. As a result, the Synth Classic Metamodel inadvertently over-represents these dominant groups and under-exploits the diversity offered by independent modellers.

Second, because many highly-rewarded miners submit essentially the same paths, the pooled metamodel distribution has insufficient variance for practical trading applications: the ensemble is concentrated around a narrow range of scenarios.

These findings led us to develop a data-driven reweighting scheme—the **Synth Weighted Metamodel**—in which miner weights are determined by each miner’s ability to forecast *hourly realized variance*, optimised via the QLIKE loss function (Section 3).

1.4 Paper Outline

The remainder of this paper is organised as follows. Section 2 presents the operator concentration analysis. Section 3 introduces the Synth Weighted Metamodel and reports backtest results. Section 4 evaluates distributional calibration through a PIT analysis. Section 5 assesses performance in simulated prediction-market experiments. Section 6 discusses implications for miners, traders, and the evolution of the SYNTH network.

2 Operator Concentration in the Synth Network

2.1 Methods

To understand the de-facto diversity of the SYNTH miner pool, we designed a clustering methodology based on summary statistics of submitted price paths.

2.1.1 Summary Statistics

For each prompt and each miner, we computed the following statistics of the submitted paths (after converting prices to log-returns):

- **Hourly volatility profile:** the realised volatility of simulated paths within each of the 24 intra-day hours.
- **Kurtosis and skewness:** tail shape and asymmetry of the 24-hour (or 60-minute) return distribution across all paths.
- **Drift:** the average return implied by the simulated paths.
- **Squared-return autocorrelation:** autocorrelations of squared returns at lags $\{1, 2, 3, 5, 10, 15, 20, 25\}$ minutes, capturing volatility clustering patterns.
- **Volatility path spread:** a measure of the dispersion of per-path volatility estimates, reflecting uncertainty modelling behaviour.
- **CRPS score:** the miner’s raw CRPS performance at the prompt, as a proxy for forecast quality.

All statistics were standardised (zero mean, unit variance) before distance computation to ensure comparability across features.

2.1.2 Pairwise Distance Metric

We operationalised the dissimilarity between two axons a and b using a *Normalised Median Absolute Deviation* (NMAD) distance, inspired by the robustness properties of the MAD (Leys et al., 2013).

For each prompt t at which both axons submitted valid predictions, we first computed the mean absolute deviation across all F standardised features:

$$\delta_{ab}(t) = \frac{1}{F} \sum_{f=1}^F |\tilde{x}_{a,f,t} - \tilde{x}_{b,f,t}|, \quad (3)$$

where $\tilde{x}_{a,f,t}$ denotes the standardised value of feature f for axon a at prompt t . The mean (rather than the median) is taken across features so that no single large feature discrepancy can mask overall similarity.

The overall pairwise distance between axons a and b is then the *median* across all prompts t at which both axons participated:

$$d(a, b) = \text{median}_t \delta_{ab}(t). \quad (4)$$

The median is chosen to be robust against isolated prompts where a miner experiences operational issues or submits atypical paths, which could spuriously inflate the apparent distance between otherwise identical operators.

2.1.3 Hierarchical Clustering

The $N \times N$ pairwise distance matrix $\mathbf{D}$ was used to construct a hierarchical clustering using *complete linkage* (Hastie et al., 2009):

$$d_{\text{complete}}(A, B) = \max_{a \in A, b \in B} d(a, b), \quad (5)$$

which enforces that every pair of axons within a cluster satisfies the distance constraint. The dendrogram was cut at a threshold $d_{\text{cut}} = 1.0$, creating groups of axons whose pairwise NMAD distances are all below this value. Singletons (axons that do not merge with any other axon at the chosen threshold) are treated as isolated nodes rather than assigned to any cluster.

The analysis was conducted over a 60-day window ending 21 March 2026, covering 9 assets in the low-frequency track.

2.2 Results

2.2.1 Distance Matrix and Network Structure

Figure 1 shows the NMAD distance matrix for the low-frequency track, with axons sorted to bring low-distance pairs close together along the diagonal. Several dense off-diagonal blocks of near-zero distances are clearly visible, indicating large groups of axons that submit nearly identical summary statistics across the observation window. This pattern is consistent with multiple miners being operated by the same entity using replicated or near-identical modelling pipelines.

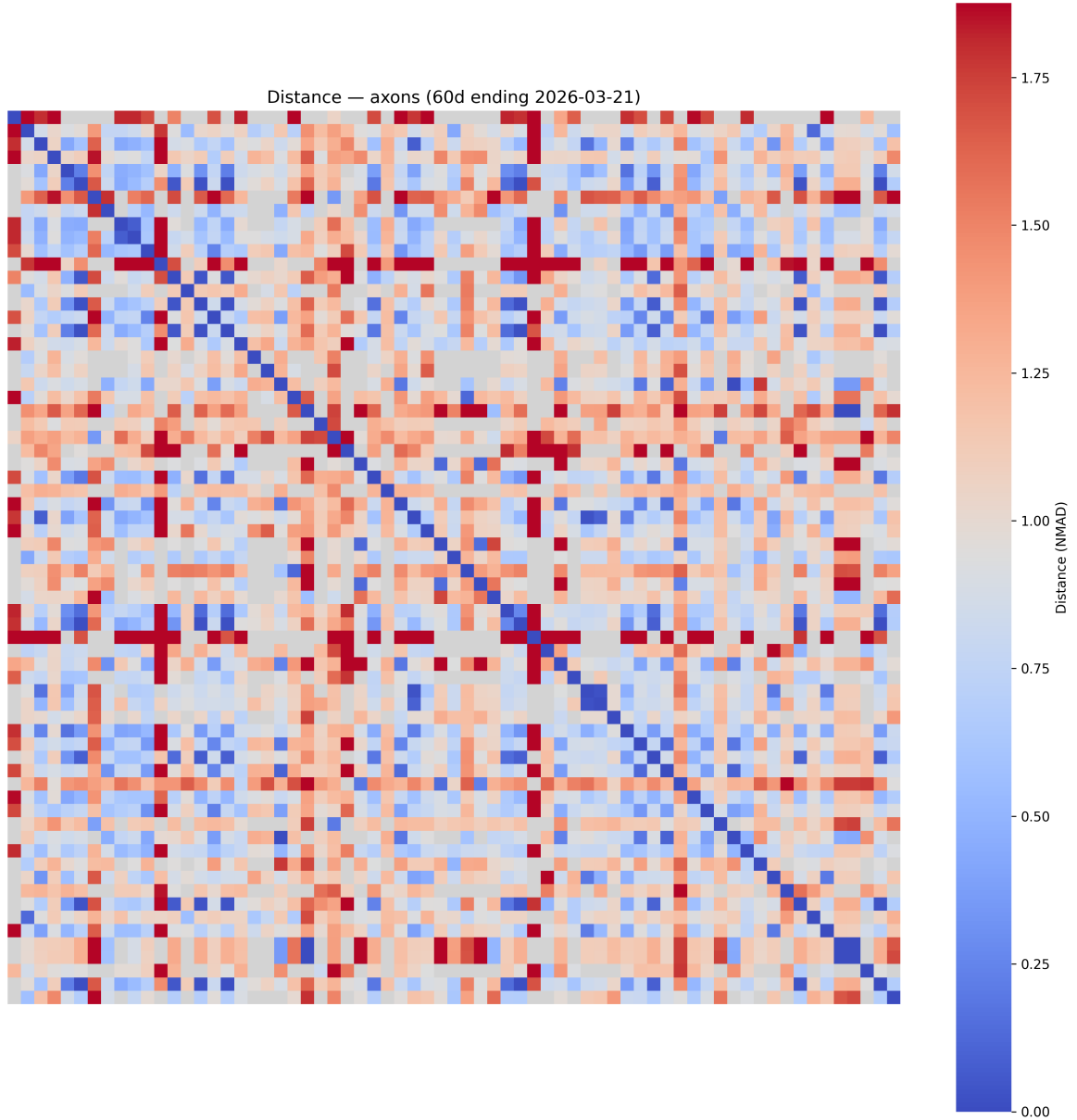


Figure 1: NMAD distance matrix for the 24-hour competition (60-day window, 21 March 2026). Lighter/blue colours indicate lower distance (higher similarity). Prominent off-diagonal blocks reveal groups of axons with nearly identical prediction patterns, pointing to coordinated multi-miner operators.

Figure 2 displays the corresponding network graph, in which nodes represent axons and edges connect pairs with NMAD below the clustering threshold. The graph has a clearly modular structure: a small number of tightly connected components (clusters) emerge, with the largest component containing **13 axons**. Nodes not connected to any cluster (singletons) are predominantly independent operators with distinct forecasting styles.

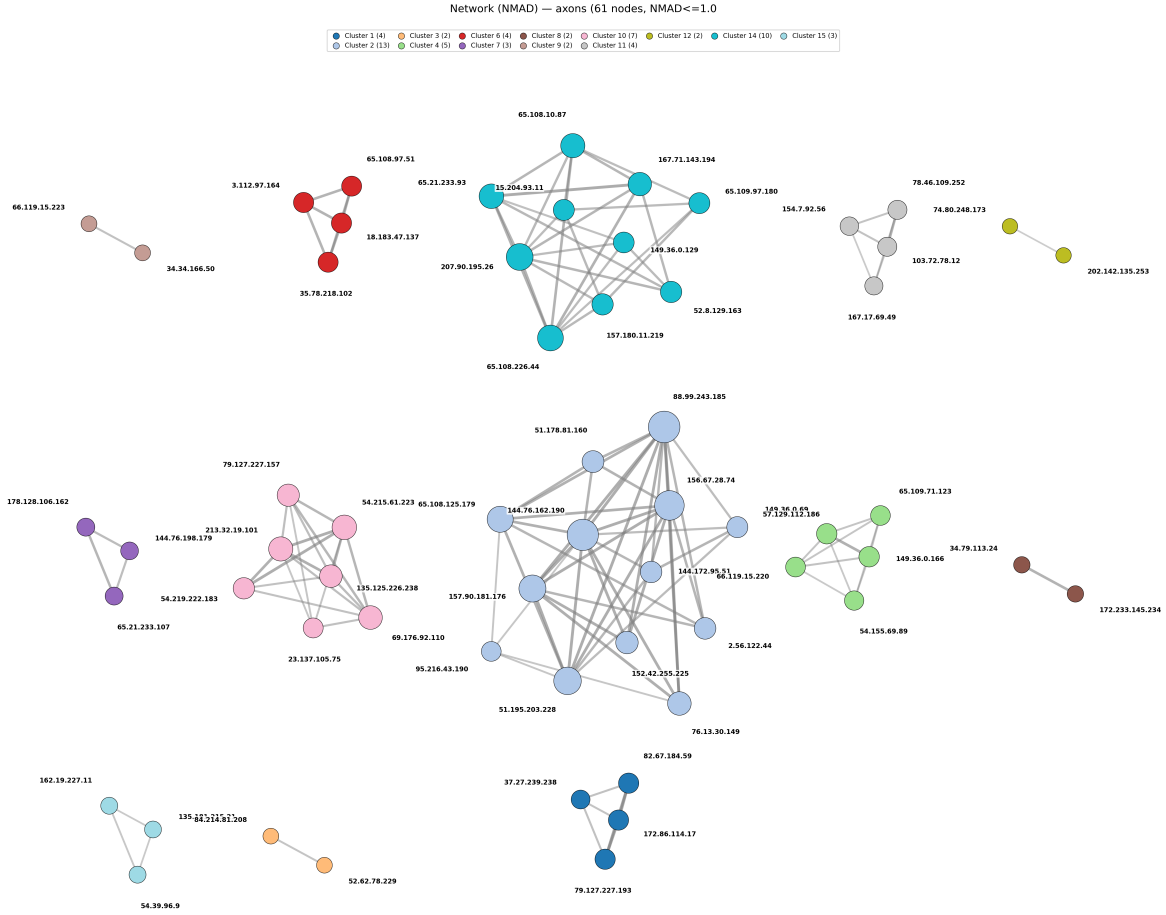


Figure 2: Network graph of axons in the 24-hour competition. Each node is an axon; edges connect pairs whose pairwise NMAD falls below the clustering threshold ($d_{\text{cut}} = 1.0$). Only within-cluster edges are shown. The largest cluster contains 13 axons—all operated by the same entity with highly consistent forecasting behaviour.

2.2.2 Cluster Profiles

Figure 3 shows the feature-level profiles of the identified clusters. Each radar panel summarises the median standardised value across all prompts and all axons within that cluster, for each feature group. Clusters differ markedly in their volatility profile, kurtosis, and autocorrelation structure, confirming that the identified groups represent genuinely distinct modelling styles—not merely noise-induced artefacts. For instance, one prominent cluster exhibits an elevated high-hour volatility profile and above-average kurtosis, consistent with a model that places more weight on end-of-day volatility and fat tails. Another cluster, by contrast, shows a flat volatility profile and low kurtosis, resembling a simple Geometric Brownian Motion (GBM) approach.

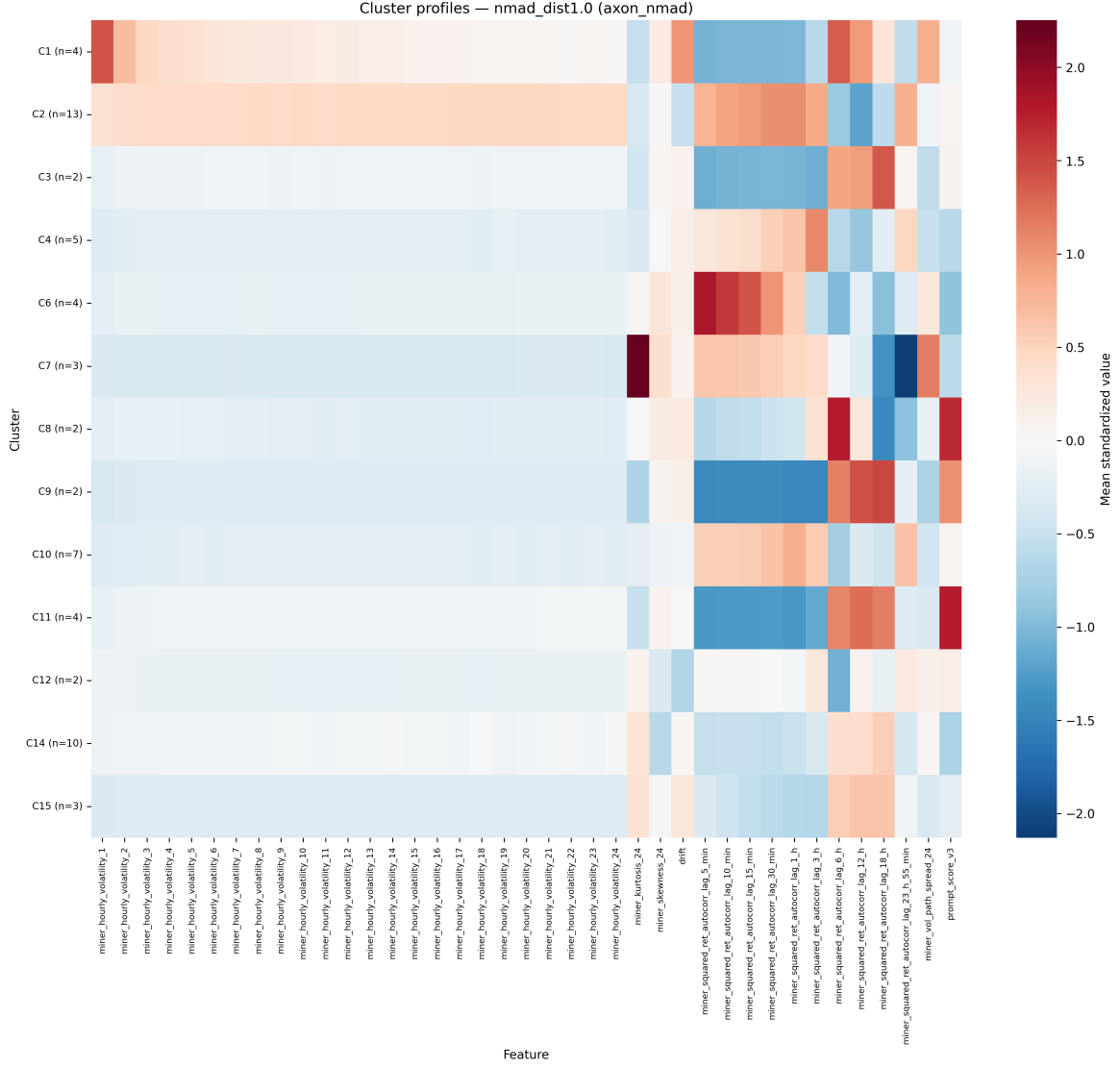


Figure 3: Feature profiles of identified clusters in the 24-hour competition. Each panel shows a radar chart of median standardised feature values for one cluster. Clusters differ substantially in volatility structure, tail behaviour, and autocorrelation, indicating that the clustering has identified truly distinct modelling approaches.

2.2.3 Reward Concentration by Cluster

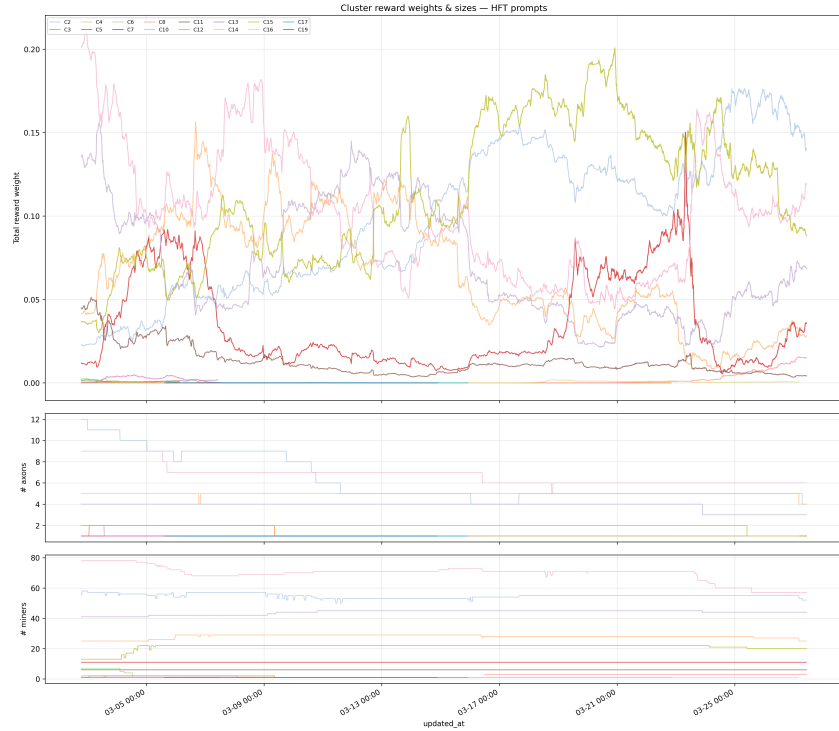
Figures 4a and 4b show the reward-weight time series disaggregated by cluster for the low-frequency and high-frequency tracks, respectively. The colour of each band identifies the cluster, while the band width at each time point reflects both the reward weight and the cluster size.

In the **low-frequency track** (Figure 4a), a handful of large clusters consistently absorb the majority of rewards throughout the observation window. The dominance is particularly pronounced after the mid-March network update, where the top two or three clusters collectively account for the bulk of total emissions—the most concentrated reward distribution observed across either competition. In the **HFT track** (Figure 4b), the reward weight is more broadly distributed across multiple players, with no single cluster dominating as heavily

as in the low-frequency competition.



(a) 24-hour competition.



(b) HFT competition.

Figure 4: Cluster reward weights over time. Band width reflects cluster size. The 24-hour competition shows the highest reward concentration, with a small number of clusters dominating emissions. The HFT competition distributes reward weight more broadly across multiple players.

Figure 5 completes the picture with a cumulative rewards curve sorted by individual miner rank. The steep initial rise confirms a highly unequal distribution: the top few axons (all belonging to one or two clusters) account for a disproportionately large share of total rewards.

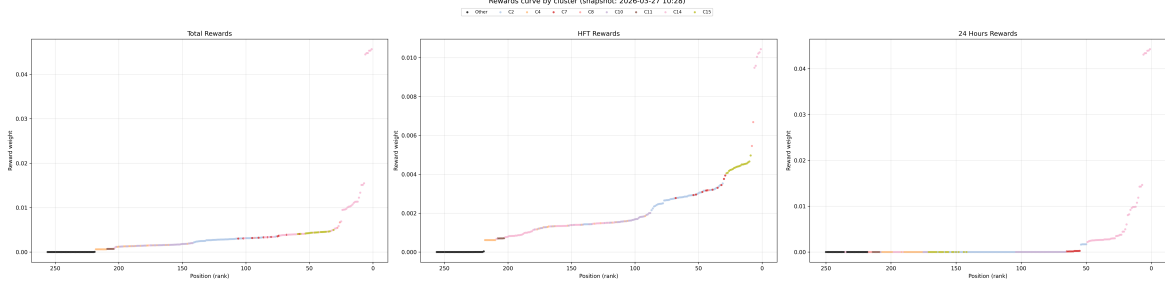


Figure 5: Cumulative reward curve by cluster. Miners are sorted by decreasing reward weight and coloured by their cluster assignment. The steep concave shape confirms heavy reward concentration among a small elite of axons belonging to the dominant clusters.

2.2.4 Implications for the Classic Metamodel

These findings have a direct bearing on the Synth Classic Metamodel. Because the top miners by cumulative reward are overwhelmingly drawn from one or two dominant clusters, the pooled metamodel is effectively an average of near-identical paths. The resulting empirical distribution inherits the stylistic properties of those dominant operators—potentially missing the diversity of approach offered by independent, well-performing miners from other clusters. This motivates a metamodel design that explicitly accounts for operator identity when allocating ensemble weight.

It should be stressed that the Synth Classic Metamodel was, and remains, a functional product: its reward-based selection is well-motivated, and the dominant clusters consistently achieve the best CRPS scores. The purpose of this section is not to criticise the existing approach but to quantify the structural limitations that motivated a more principled redesign.

3 The Synth Weighted Metamodel

3.1 Methods

3.1.1 Design Philosophy

The Synth Weighted Metamodel departs from the heuristic reward-based selection of its predecessor in two respects:

1. Weights are determined by a *forecasting criterion* grounded in statistical theory—specifically, by the ability of each miner to predict the realized hourly variance of asset returns.
2. The optimisation procedure implicitly discourages over-concentration on near-identical miners, since correlated miners provide redundant information with diminishing marginal value.

We deliberately omit a full description of the internal optimisation procedure so as not to expose proprietary implementation details.

3.1.2 Variance Forecasting and the QLIKE Loss

The central idea is that a well-calibrated price path should imply an accurate forecast of future realized variance. For a given prompt, each miner’s submitted paths $\{p_{i,t}\}$ are used to estimate an hourly variance forecast $\hat{\sigma}_{i,h}^2$ for each hour h ahead. The quality of this forecast is evaluated against the corresponding realized hourly variance σ_h^2 using the *Quasi-Likelihood* (QLIKE) loss function (Patton, 2011):

$$\mathcal{L}_{\text{QLIKE}}(\hat{\sigma}^2, \sigma^2) = \log \hat{\sigma}^2 + \frac{\sigma^2}{\hat{\sigma}^2}. \quad (6)$$

The QLIKE loss has several properties that make it the ideal choice for variance forecast evaluation (Patton, 2011):

- It is a *consistent* loss function: when the realized variance proxy is noisy (as it inevitably is with discretely sampled data), minimising QLIKE in expectation still yields the true conditional variance as the optimal forecast. By contrast, the commonly used MSE loss is consistent only when the proxy is unbiased and noise-free.
- It imposes an *asymmetric* penalty: underestimating variance is punished more severely than overestimating it. This is desirable from a risk-management perspective, where under-coverage of uncertainty is typically more costly than over-coverage.
- The term $\sigma^2/\hat{\sigma}^2$ diverges as $\hat{\sigma}^2 \rightarrow 0$, providing a strong deterrent against degenerate forecasts that collapse the predicted distribution to a point mass.
- It is related to the negative log-likelihood under a $\text{Gamma}(1, \sigma^2)$ observation model, giving it a natural probabilistic interpretation.

The ensemble variance forecast is the weighted sum of individual miner forecasts:

$$\hat{\sigma}_h^2(\mathbf{w}) = \sum_{i=1}^N w_i \hat{\sigma}_{i,h}^2, \quad \sum_{i=1}^N w_i = 1, \quad w_i \geq 0. \quad (7)$$

The miner weights $\mathbf{w}$ are optimised to minimise the average QLIKE loss over a rolling in-sample window:

$$\mathbf{w}^* = \arg \min_{\mathbf{w}} \frac{1}{T} \sum_{t=1}^T \sum_h \mathcal{L}_{\text{QLIKE}}(\hat{\sigma}_h^2(\mathbf{w}), \sigma_{h,t}^2), \quad (8)$$

subject to the simplex constraint $\mathbf{w} \in \Delta^{N-1}$.

3.1.3 Backtest Setup

We conducted backtests over the following periods:

- **24-hour (LF) competition:** approximately two months from late January to mid-March 2026.

- **HFT competition:** approximately one month (chosen to be shorter owing to the higher frequency of prompts and the consequently larger data volume).

At each evaluation point, weights were estimated on in-sample data and applied to out-of-sample paths. Performance is reported separately for in-sample (IS) and out-of-sample (OOS) periods.

3.2 Results

3.2.1 MSE Performance

Table 1 reports the out-of-sample MSE of the variance forecasts. Tables 2 and 3 compare key weight-concentration statistics; for the classic metamodel, these correspond to the behaviour of the *current weighting system* (the reward-based pooling scheme).

Table 1: Out-of-sample MSE of forecasted variances — both competitions. All values rounded to three decimal places.

Model	OOS MSE (24h)	OOS MSE (HFT)
Synth Classic Metamodel	0.932	0.095
Synth Weighted Metamodel	0.920	0.082

Table 2: Weight-concentration statistics — 24-hour competition. Averages over all backtest evaluation points. For the classic model these describe the *current weighting system*. All values rounded to three decimal places.

Weighting Scheme	Mean Entropy	Mean Max Weight	Mean Active Miners
Current Weighting System	4.143	0.149	218.007
Synth Weighted Metamodel weights	2.441	0.262	117.060

Table 3: Weight-concentration statistics — HFT competition. Same metrics as Table 2.

Weighting Scheme	Mean Entropy	Mean Max Weight	Mean Active Miners
Current Weighting System	4.987	0.018	232.670
Synth Weighted Metamodel weights	3.725	0.180	212.289

The Synth Weighted Metamodel achieves a **1.2% reduction in OOS MSE** in the 24-hour competition and a **13.1% reduction** in the HFT competition. The larger improvement in the HFT track reflects the greater signal available from the higher frequency of prompts, which provides more data for the optimisation and allows the weights to adapt more rapidly to changes in miner quality.

Turning to the weight-concentration statistics: the mean Shannon entropy (Shannon, 1948)

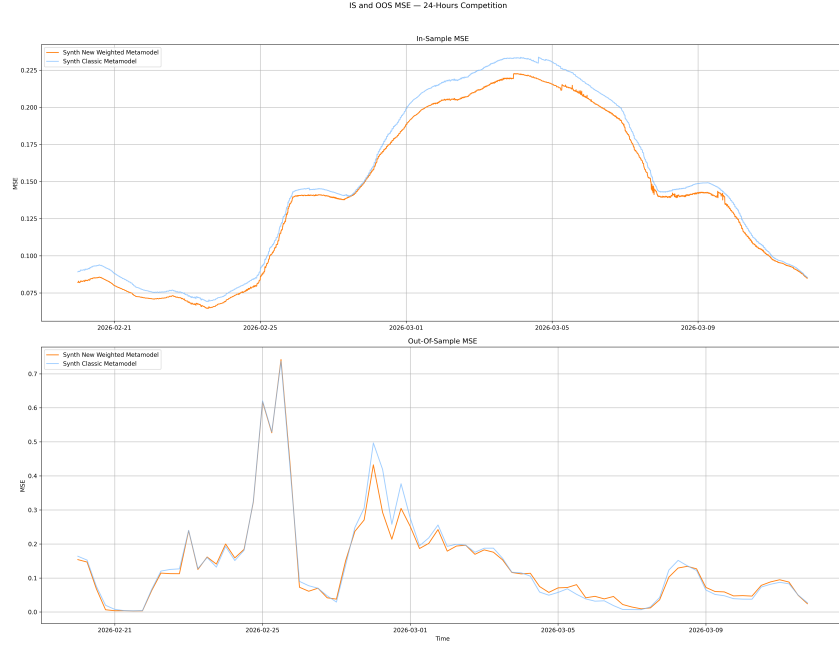
$$H(\mathbf{w}) = - \sum_{i=1}^N w_i \log w_i \quad (9)$$

is substantially lower under the new model (2.441 vs. 4.143 for LF; 3.725 vs. 4.987 for HFT), indicating that the Synth Weighted Metamodel assigns significantly more concentrated weights. Correspondingly, the mean maximum weight is higher (0.262 vs. 0.149 for LF; 0.180 vs. 0.018 for HFT), and the number of active miners receiving positive weight is lower (117.060 vs. 218.007 for LF).

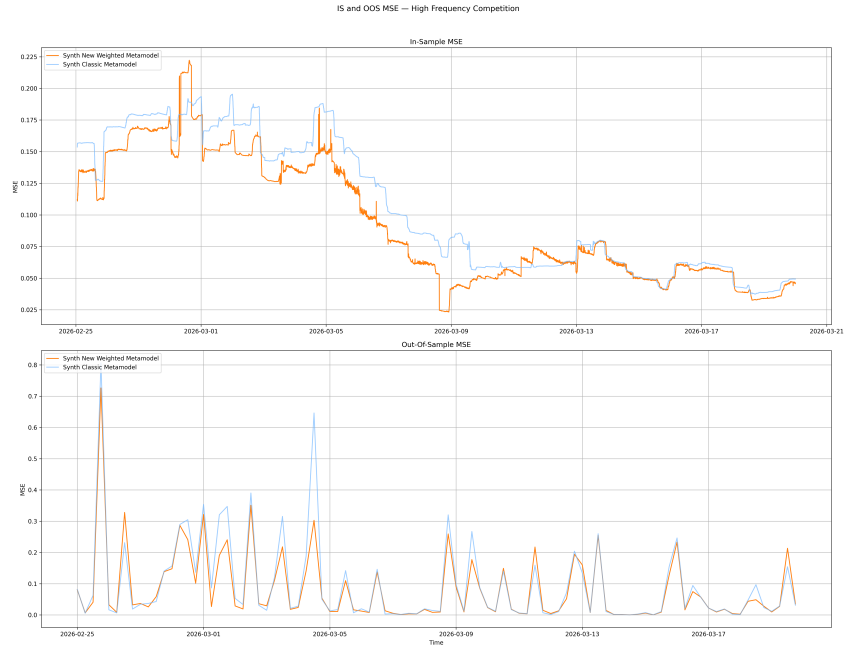
These findings are consistent with the theoretical motivation: when many miners produce correlated forecasts, a QLIKE-minimising ensemble correctly concentrates weight on the single best representative among them. The Synth Weighted Metamodel thus acts as an implicit *diversity filter*: rather than down-weighting miners with low cumulative reward, it down-weights miners whose variance forecasts add no information beyond what is already captured by higher-ranked miners.

3.2.2 IS and OOS MSE Over Time

Figures 6a and 6b show the IS and OOS MSE time series for both models in the LF and HFT competitions, respectively. The new model maintains a consistent edge over the baseline in OOS MSE across the full backtest window, with no evidence of model instability or over-fitting. The gap is most pronounced in the HFT competition, where the new model’s OOS MSE tracks closely below the classic baseline throughout.



(a) 24-hour competition.



(b) HFT competition.

Figure 6: In-sample (IS) and out-of-sample (OOS) MSE over the backtest period. The Synth Weighted Metamodel consistently achieves lower OOS MSE, with the advantage most pronounced in the HFT track.

3.2.3 Pairwise Forecast Correlations

Figures 7a and 7b show the pairwise OOS forecast correlation matrix between the Synth Classic Metamodel and the Synth Weighted Metamodel. The high correlation between the two metamodels—despite the significant difference in weighting procedure—confirms that they are both targeting the same underlying distributional signal, while the differences in

concentration and diversity lead to distinct out-of-sample performance.

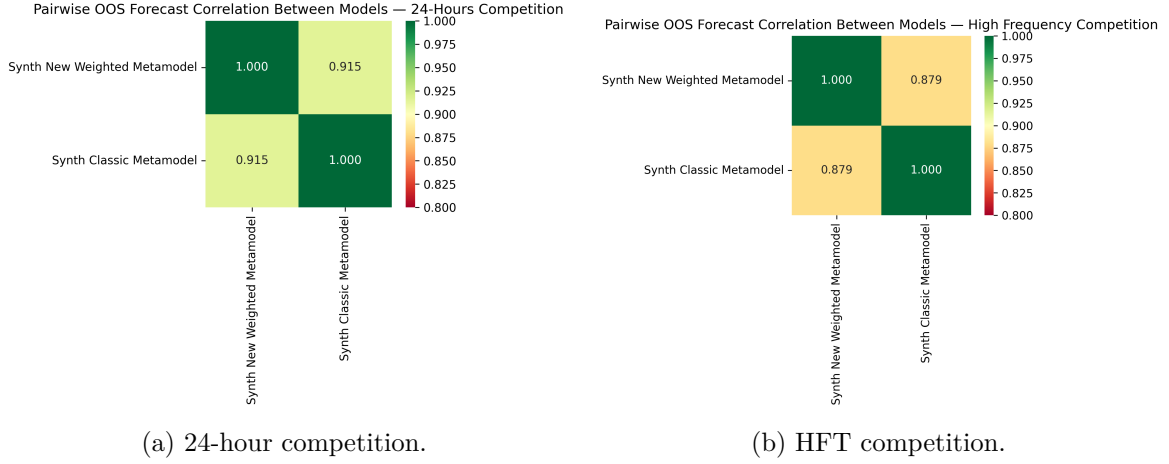
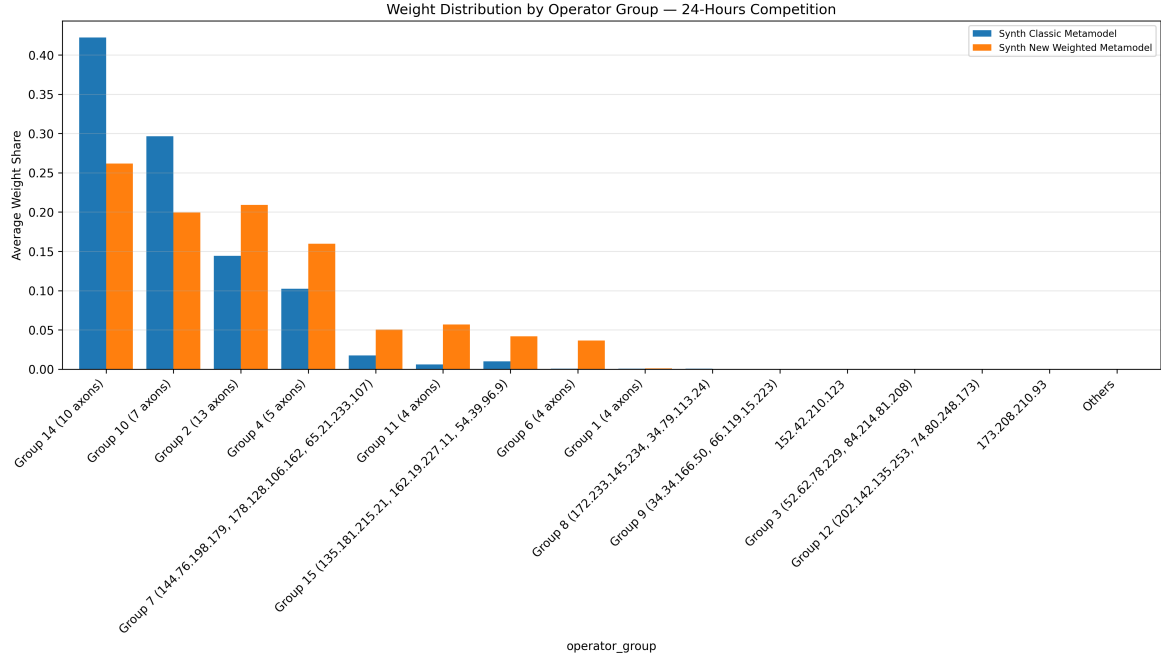


Figure 7: Pairwise OOS forecast correlation between models. Colours range from blue (low correlation) to red (high correlation). Both metamodels are closely correlated with each other, confirming they target the same underlying signal despite their different weighting procedures.

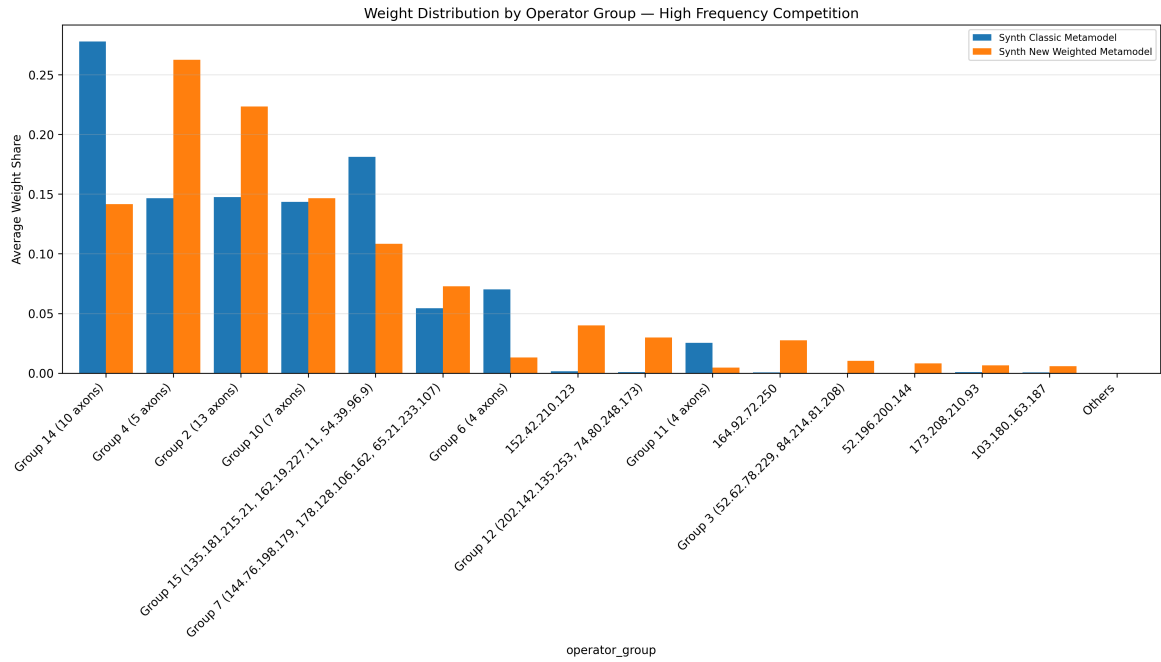
3.2.4 Weight Distribution by Operator Group

Figures 8a and 8b compare the weight distribution across operator clusters (identified in Section 2) under both weighting schemes. Each boxplot shows the distribution of weights assigned to miners belonging to each identified cluster.

Under the **current weighting system**, miners from all clusters receive broadly similar weights, since all top-rewarded miners are pooled with equal weight regardless of operator identity. Under the **Synth Weighted Metamodel**, weight is more selective but distributed across miners representing several *independent modelling styles*, rather than concentrating on a single dominant cluster. This is precisely the diversity property the design aimed to achieve: forecasting quality drives selection, not merely cumulative reward.



(a) 24-hour competition.



(b) HFT competition.

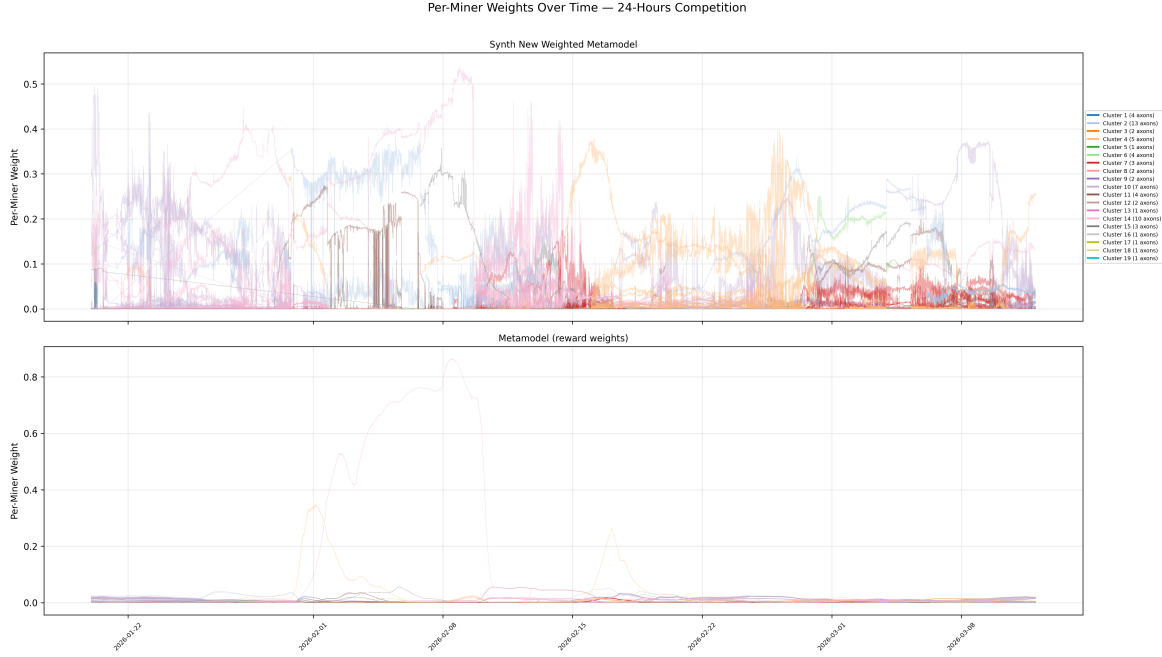
Figure 8: Distribution of miner weights by operator cluster under both weighting schemes. Under the new model, weight is more selective but distributed across multiple independent operators, improving forecast diversity.

3.2.5 Per-Miner Weight Trajectories

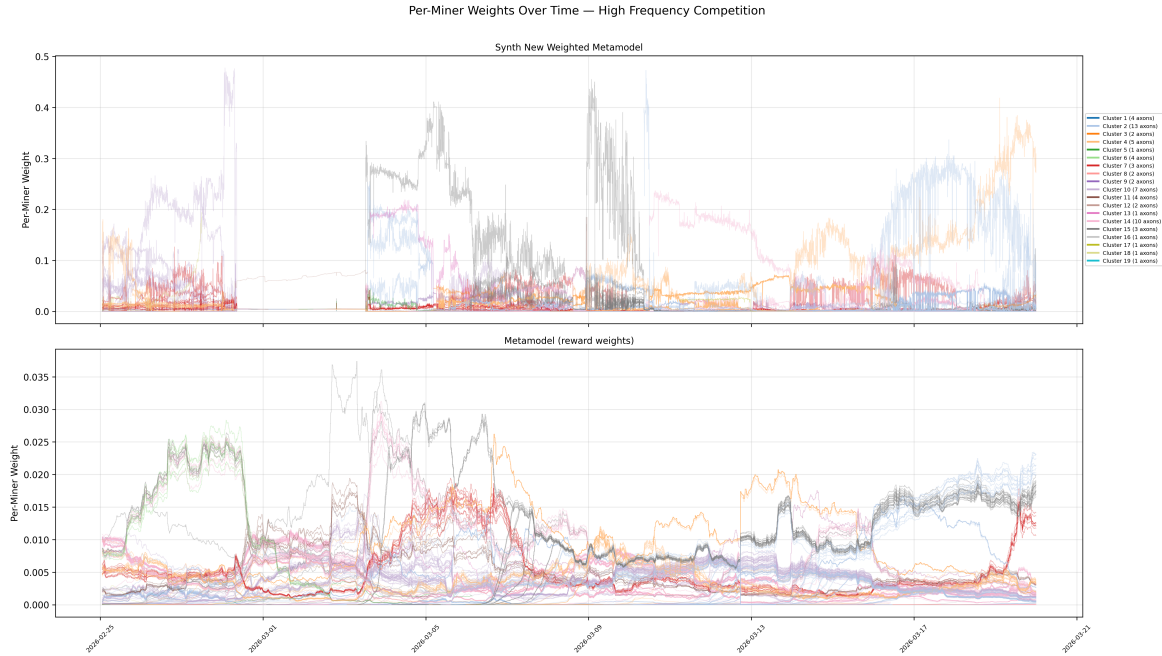
Figures 9a and 9b show the evolution of per-miner weights over the backtest period, coloured by cluster membership. A striking feature of the Synth Weighted Metamodel's weight trajectories is that, within each cluster, it is typically *only one miner* that receives substantial weight at any given time step. This is consistent with the theoretical property of QLIKE

minimisation: when multiple miners produce correlated forecasts, the optimiser concentrates on the best single representative, discarding redundant copies.

In the HFT competition (Figure 9b), the contrast with the current weighting system is particularly stark: whereas the current system distributes weight broadly across many miners from the same cluster, the Synth Weighted Metamodel cleanly identifies and rewards the single best performer within each operator group at each point in time.



(a) 24-hour competition.



(b) HFT competition.

Figure 9: Per-miner weight trajectories over the backtest period, coloured by cluster. Under the Synth Weighted Metamodel (bottom panel of each pair), typically only one miner per cluster receives meaningful weight—the clearest manifestation of the diversity objective.

4 Distributional Calibration: PIT Analysis

4.1 Methods

A key requirement for any ensemble forecast used in trading applications is *distributional calibration*: the empirical distribution of predicted price changes should match the true distribution of realized outcomes. We assess calibration using the *Probability Integral Transform* (PIT; Diebold et al., 1998; Rosenblatt, 1952).

4.1.1 PIT Definition

At each evaluation tick τ , for a given forward horizon h minutes, let $\hat{F}_\tau$ denote the empirical CDF of predicted price changes implied by the metamodel at tick τ , and let $y_{\tau,h}$ denote the realized price change from τ to $\tau + h$ minutes. The PIT value is:

$$u_{\tau,h} = \hat{F}_\tau(y_{\tau,h}) = \Pr(\Delta\hat{p} < y_{\tau,h}), \quad (10)$$

where the probability is computed over the ensemble of predicted price changes.

For a *weighted* metamodel, the CDF is the mixture:

$$\hat{F}_\tau(y) = \sum_{i=1}^N w_i \hat{F}_{i,\tau}(y), \quad (11)$$

where $\hat{F}_{i,\tau}$ is miner i 's empirical CDF and w_i is the assigned weight.

Definition 4.1 (Calibration). A forecasting model is *calibrated* if and only if the PIT values $\{u_{\tau,h}\}$ are uniformly distributed on $[0, 1]$.

Departures from uniformity carry the following interpretations:

- **U-shaped histogram**: under-dispersed forecast (too narrow predicted distribution; model is over-confident).
- **Hump in the middle**: over-dispersed forecast (predicted distribution too wide; model is under-confident).
- **Left skew**: model systematically predicts prices too high.
- **Right skew**: model systematically predicts prices too low.

4.1.2 Kolmogorov–Smirnov Test

We quantify the deviation from uniformity using the two-sided *Kolmogorov–Smirnov* (KS) statistic (Kolmogorov, 1933; Smirnov, 1948):

$$D_n = \sup_{u \in [0,1]} |\hat{G}_n(u) - u|, \quad (12)$$

where $\hat{G}_n$ is the empirical CDF of the n PIT values. The KS statistic ranges from 0 (perfect uniformity) to 1 (complete misalignment); smaller values indicate better calibration.

4.1.3 Additional Diagnostics

Alongside PIT histograms and KS statistics, we report:

- **QQ plots:** quantile–quantile plots of PIT values against the theoretical Uniform(0,1) distribution. Perfect calibration corresponds to a 45° diagonal.
- **Reliability diagrams:** binned calibration plots comparing predicted event probabilities to observed frequencies.
- **Tail analysis:** PIT values restricted to the lower and upper deciles, assessing whether tail risks (extreme price moves) are captured accurately.

The analysis was conducted for all out-of-sample evaluation points in the backtest, evaluated at horizons $h \in \{5, 15, 60, 240\}$ minutes (LF) and $h \in \{1, 5, 15, 60\}$ minutes (HFT). A GBM baseline (estimated from recent realized volatility) is included in all figures as a benchmark representing the calibration of a standard parametric model. All results are out-of-sample.

4.2 Results

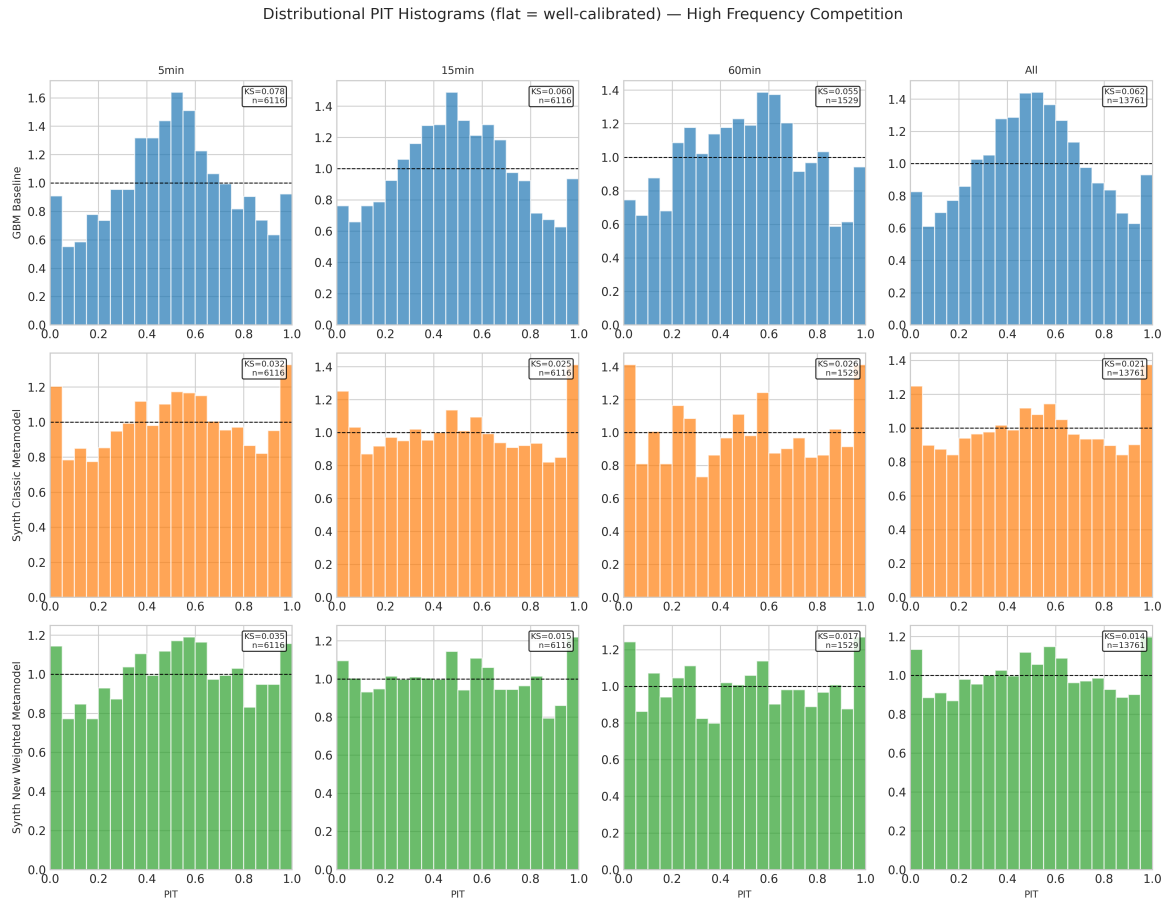
4.2.1 PIT Histograms

Figures 10a and 10b show the PIT histograms for all three models (GBM, Synth Classic Metamodel, Synth Weighted Metamodel) and all evaluation horizons. Each panel reports the KS statistic in the figure legend.

In both the **24-hour** and the **HFT** competitions, the Synth Weighted Metamodel achieves the lowest KS statistic across all horizons, indicating the closest adherence to the desired uniform distribution. The GBM baseline exhibits a noticeable hump-shaped pattern at short horizons, suggesting over-dispersion typical of a constant-volatility model that fails to account for intra-day volatility clustering. The Synth Classic Metamodel falls between GBM and the new model, with somewhat better calibration at shorter horizons than GBM, but still showing systematic deviations that the Synth Weighted Metamodel corrects.



(a) 24-hour competition.

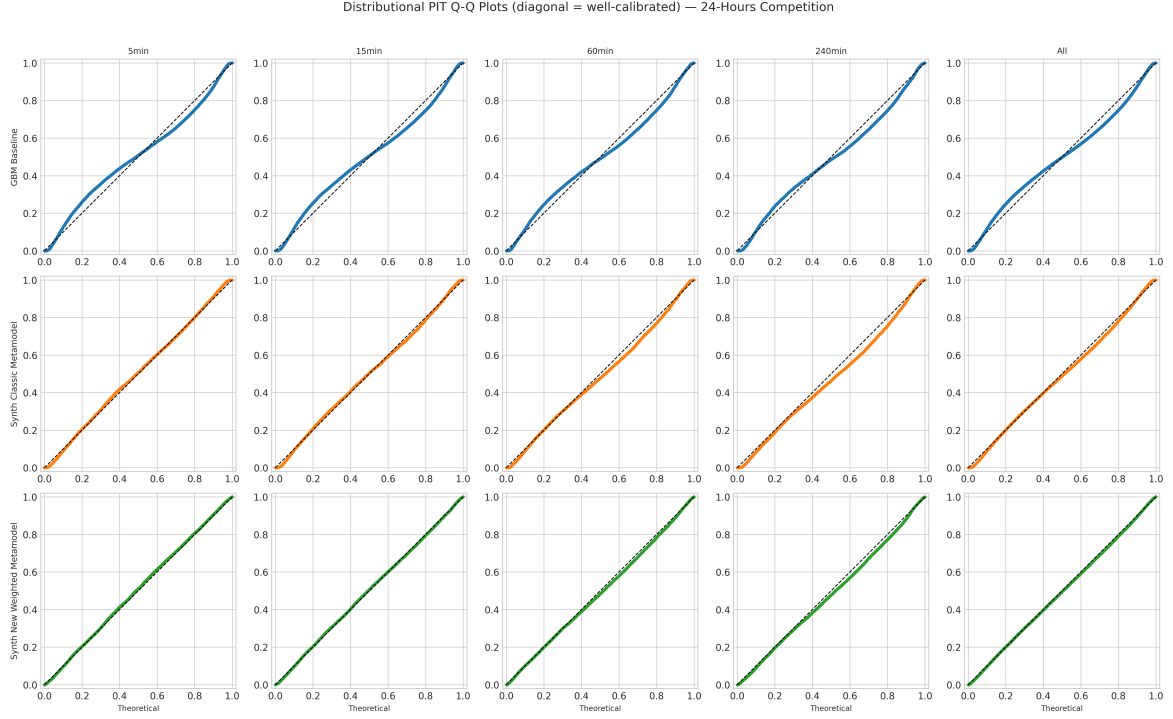


(b) HFT competition.

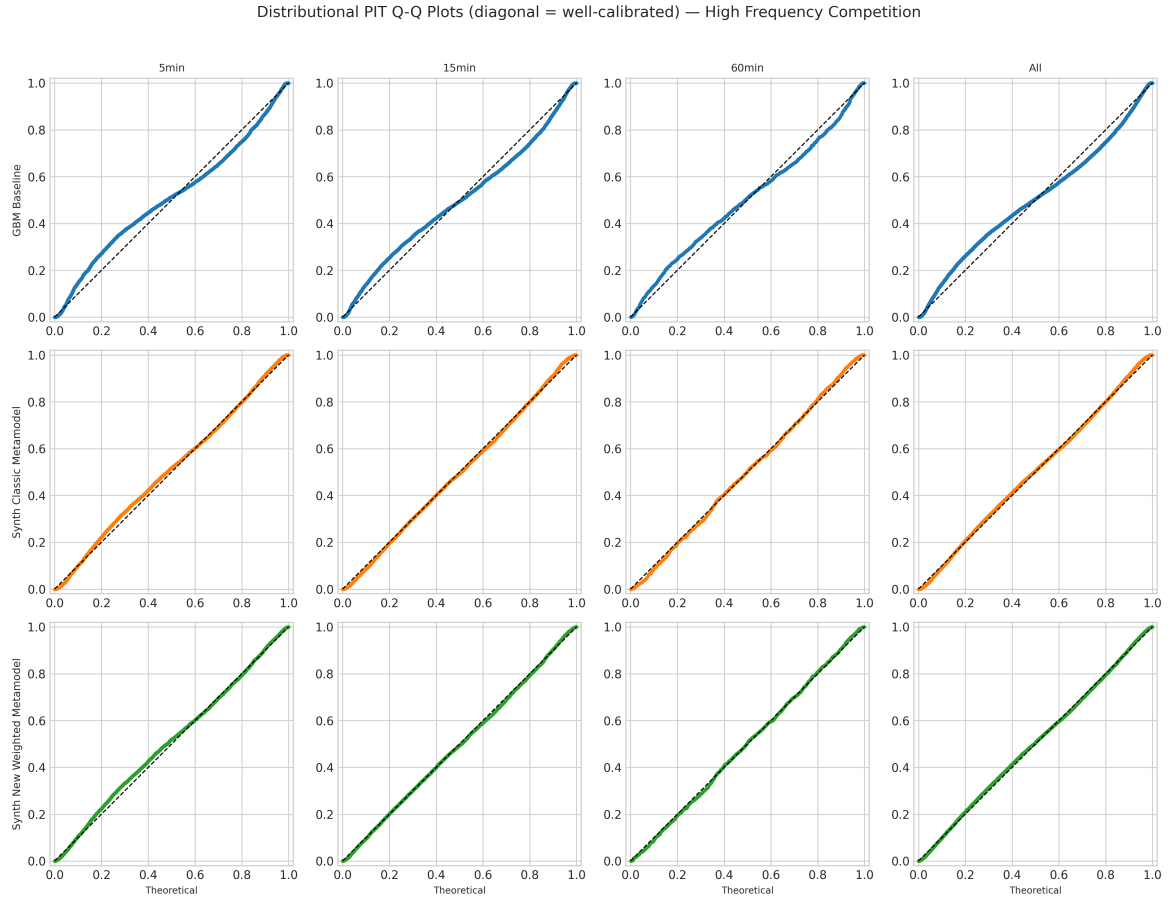
Figure 10: PIT histograms for all models. Each panel corresponds to a different evaluation horizon. KS statistics (reported in legends) are lower for the Synth Weighted Metamodel across all horizons in both competitions, indicating superior calibration.

4.2.2 QQ Plots

Figures 11a and 11b show the corresponding QQ plots. Points on the 45° diagonal indicate a perfectly calibrated model. The Synth Weighted Metamodel tracks the diagonal most closely, particularly in the central quantile range. Notable deviations at the extreme tails ($u < 0.05$ and $u > 0.95$) are present across all models and reflect the inherent difficulty of predicting extreme market moves—a challenge that no purely data-driven model can fully resolve.



(a) 24-hour competition.

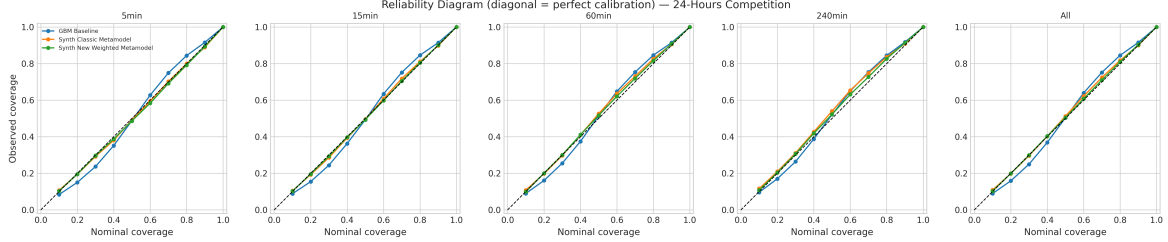


(b) HFT competition.

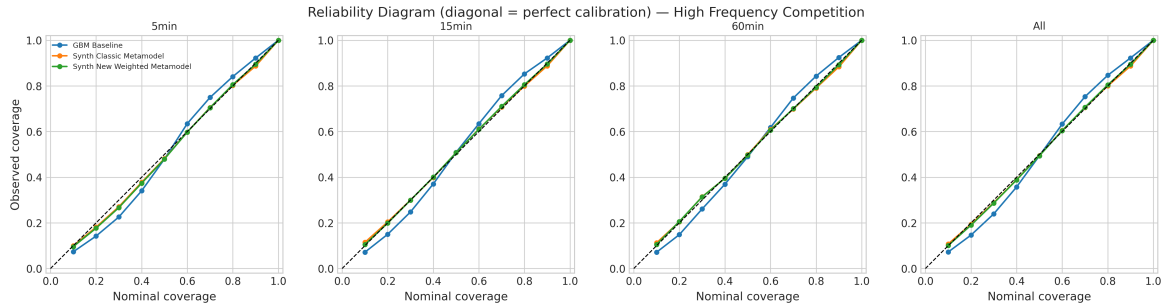
Figure 11: QQ plots of PIT values against the theoretical $\text{Uniform}(0,1)$ distribution. Deviations from the diagonal indicate miscalibration. The Synth Weighted Metamodel lies closest to the diagonal across both competitions.

4.2.3 Reliability Diagrams

Figures 12a and 12b show reliability diagrams, which assess whether predicted event probabilities correspond to observed frequencies. For a perfectly calibrated model, the diagram falls on the identity line. The Synth Weighted Metamodel shows the tightest alignment to the diagonal, confirming that its probability statements accurately reflect empirical event frequencies.



(a) 24-hour competition.



(b) HFT competition.

Figure 12: Reliability diagrams. Each point represents the observed event frequency in a predicted probability bin. Deviation from the identity line measures miscalibration. The Synth Weighted Metamodel is most closely aligned with the identity line in both competitions.

4.2.4 Tail Analysis

Figures 13a and 13b focus on tail calibration: they compare the predicted probability assigned to extreme price moves (below the 5th percentile or above the 95th percentile of the predicted distribution) against the realized frequency of such events. Accurate tail coverage is critical for trading strategies that use SYNTH's forecasts for risk management or options pricing.

Across both competitions, the Synth Weighted Metamodel exhibits more accurate tail coverage than either the GBM baseline or the Synth Classic Metamodel. In particular, the GBM baseline systematically under-estimates the probability of extreme upward moves in the HFT competition—a consequence of its symmetric, static volatility assumption—while the Synth Weighted Metamodel corrects for this asymmetry.

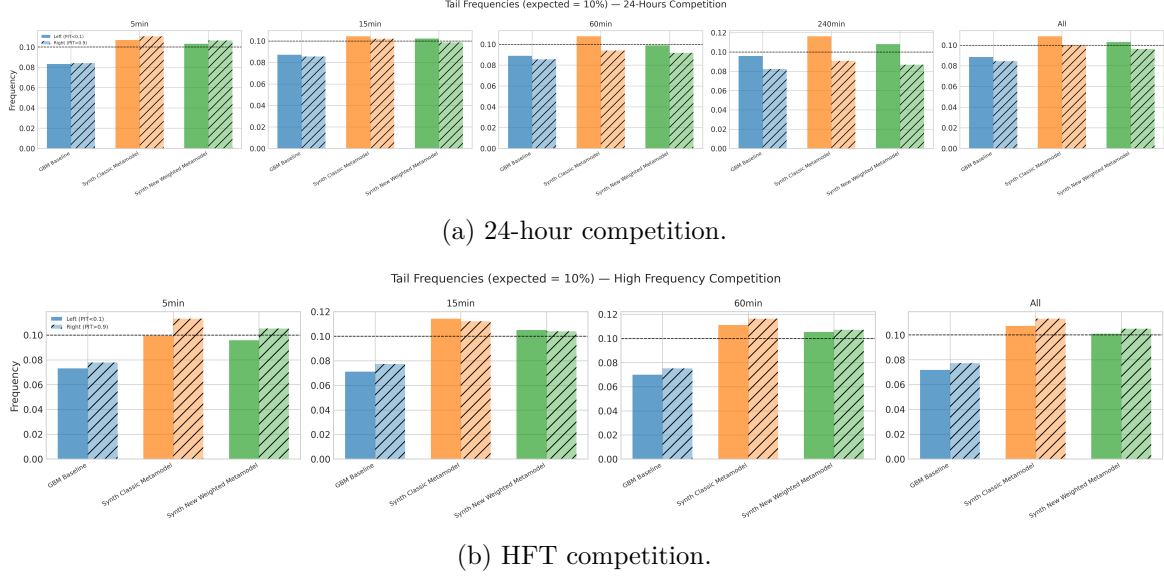


Figure 13: Tail analysis: predicted vs. realised probability of extreme price moves (below 5th and above 95th percentile of the predicted distribution). The Synth Weighted Metamodel achieves the most accurate tail coverage in both competitions.

5 Prediction Market Simulation

5.1 Methods

A natural application of probabilistic price forecasts is the construction of prediction-market probabilities. We simulate how each model would perform as a prediction-market participant by translating the ensemble price distributions into binary event probabilities and evaluating them against realized outcomes.

5.1.1 Market Simulation Setup

For each prompt, we simulate a prediction market on the event “the final asset price exceeds the price at prompt start time.” At each evaluation tick τ within the forecast horizon, the predicted probability of the event is:

$$p_\tau = \Pr(\hat{p}_T > p_0 \mid \text{paths at } \tau), \quad (13)$$

where p_0 is the starting price and $\hat{p}_T$ is the simulated terminal price. The probability is updated every minute (HFT competition) or every 5 minutes (24-hour competition) as new price observations arrive and paths are conditioned on the realized trajectory.

We ran:

- **24-hour competition:** markets evaluated at 5-minute resolution.
- **HFT 60-minute competition:** markets evaluated at 1-minute resolution.
- **HFT 15-minute competition:** markets evaluated at 1-minute resolution.

5.1.2 Scoring Rules

We evaluate prediction-market performance using two complementary metrics:

Brier Score (Brier, 1950): the mean squared error between the predicted probability and the binary realized outcome:

$$\text{BS}_\tau = (o_\tau - p_\tau)^2, \quad (14)$$

where $o_\tau \in \{0, 1\}$ is the realized binary outcome at tick τ . Lower Brier scores indicate better probabilistic forecasts; a score of 0 corresponds to perfect prediction.

Log-Likelihood: the logarithmic scoring rule, proper for binary events:

$$\text{LL}_\tau = o_\tau \log p_\tau + (1 - o_\tau) \log(1 - p_\tau), \quad (15)$$

clipped to avoid numerical issues. Higher log-likelihoods indicate better calibrated probability forecasts.

5.2 Results

5.2.1 Probability Trajectories

Figures 14–16 show representative probability trajectories for the 24-hour, HFT 60-minute, and HFT 15-minute competitions, respectively. Each line tracks the predicted probability of the target event as a function of time remaining in the market.

A key observation is that the **GBM baseline** tends to converge to the true realized probability more slowly than both metamodels, particularly in the early portion of the forecast window. This is consistent with GBM’s lack of conditioning on recent realized price movements: its probability estimates are anchored to a long-run volatility estimate and adjust sluggishly as new information arrives. Both SYNTH models update probabilities more dynamically, reflecting the richer distributional information embedded in the miner-generated paths.

The Synth Weighted Metamodel updates probabilities similarly to the Synth Classic Metamodel but with more concentrated weighting, reducing dependence on idiosyncratic miner paths and producing more stable probability estimates that are easier to trade on.

A particularly noteworthy feature is visible in the **HFT 15-minute competition** (Figure 16, middle panel on the left): near the end of the 15-minute market, all models—including GBM—appear to converge to the *wrong* outcome. This is not a model failure but an inherent property of very short-horizon markets: a sudden 1-minute price jump in the final seconds of the window can determine the binary outcome after all models have already committed to a high-probability estimate. In these edge cases, no data-driven model can “correct” in time, and the result is an episodically elevated Brier score.

Probability Trajectories — Sample Contests — 24-Hours Competition

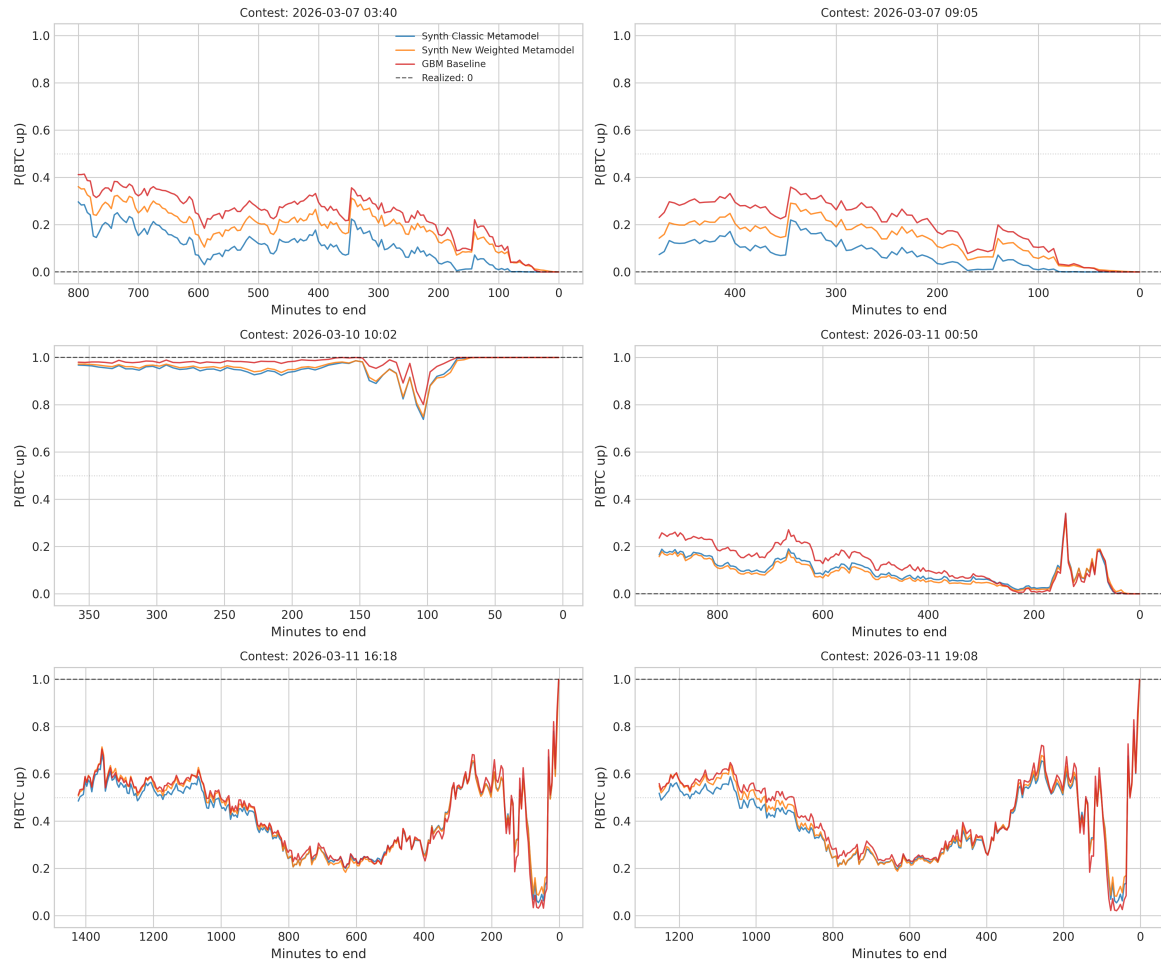


Figure 14: Representative probability trajectories — 24-hour competition. GBM converges most slowly to the true realized probability; SYNTH models respond faster to incoming price data, reflecting the richer information in miner-generated paths.

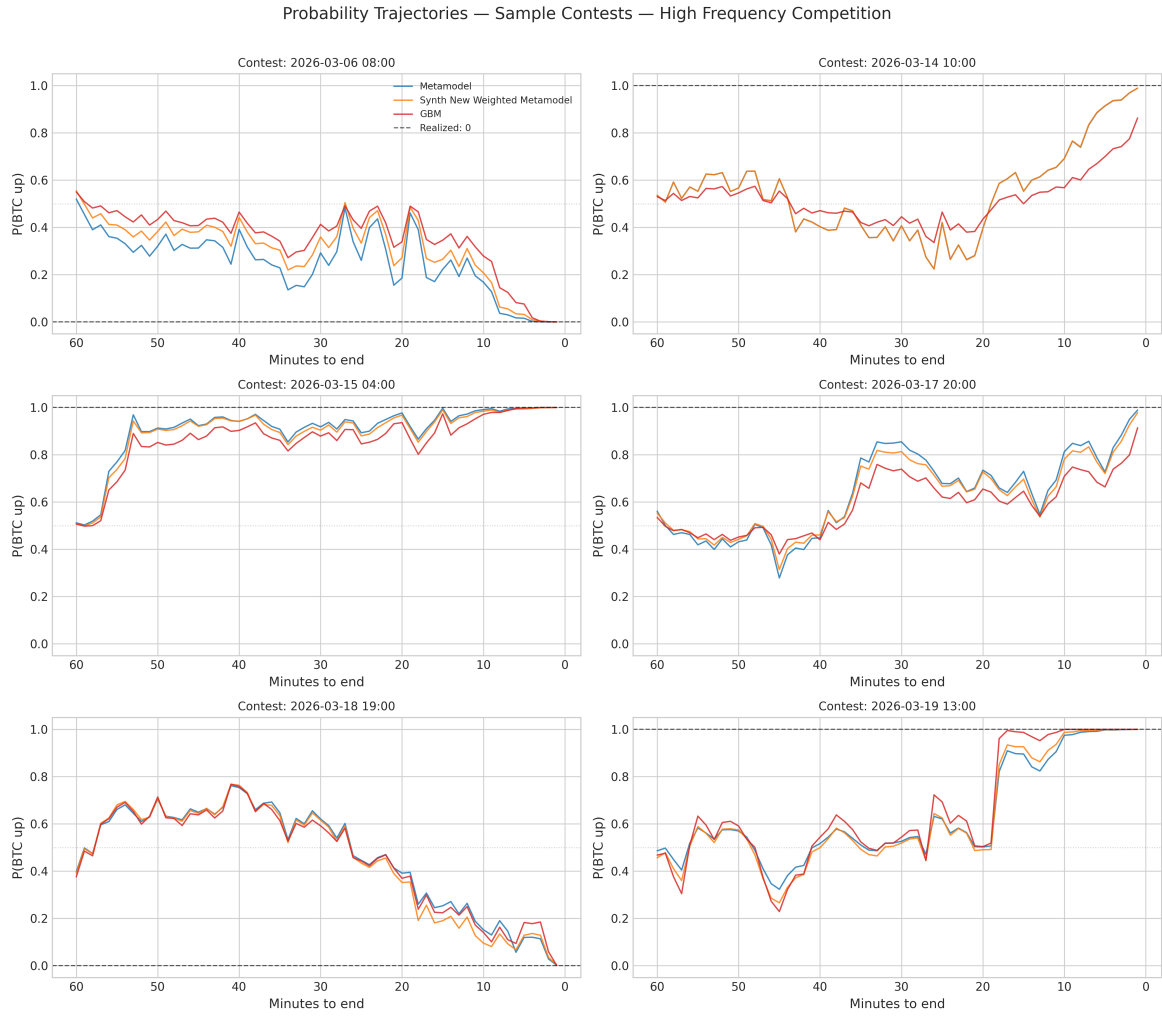


Figure 15: Representative probability trajectories — HFT 60-minute competition. Both SYNTH models update probabilities more dynamically than the GBM baseline. The Synth Weighted Metamodel tracks the Synth Classic Metamodel closely, with more concentrated weighting reducing dependence on idiosyncratic miner paths.

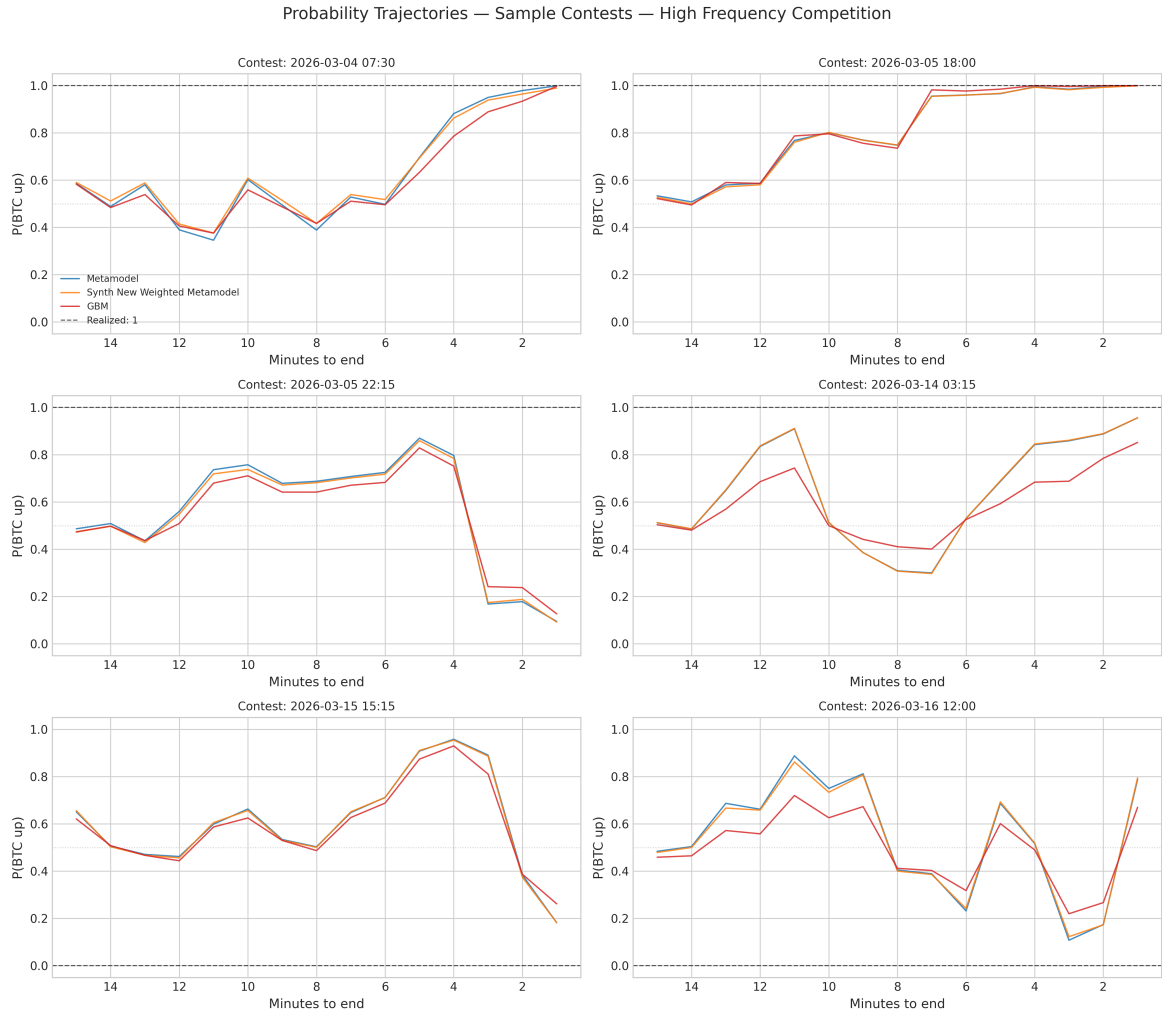
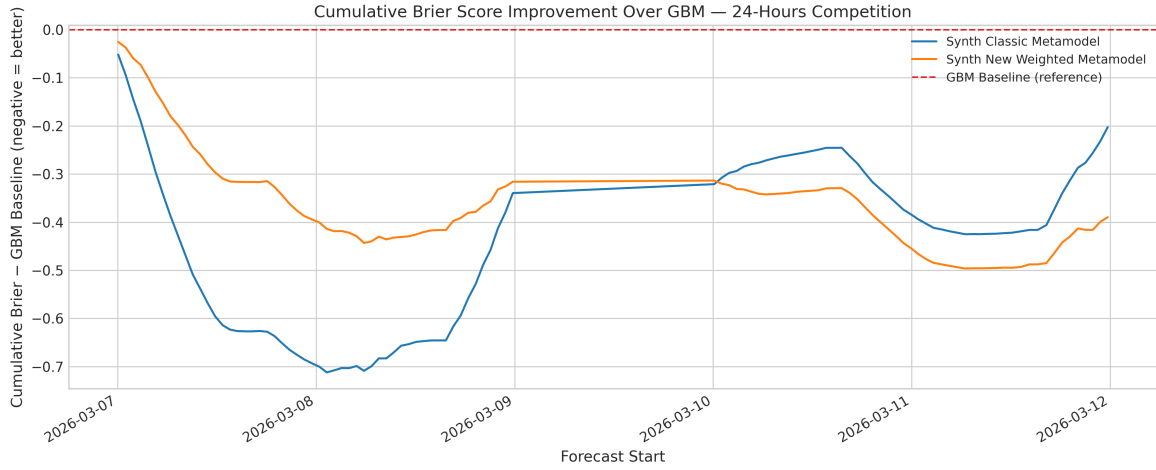


Figure 16: Representative probability trajectories — HFT 15-minute competition. In the middle panel on the left, all models briefly converge toward the wrong outcome near market close due to a last-second price jump—an inherent limitation of ultra-short-horizon markets, not a model artefact.

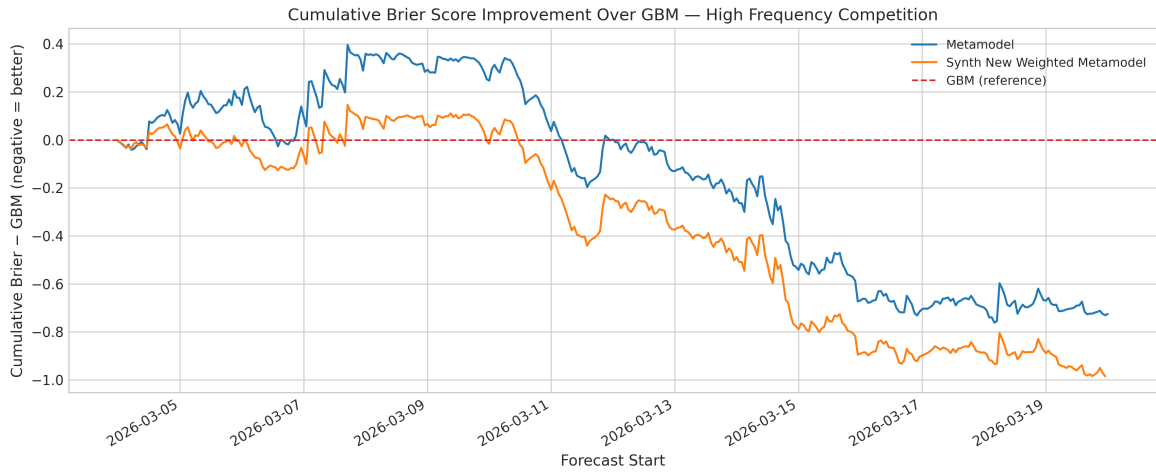
5.2.2 Cumulative Brier Scores

Figures 17a, 17b, and 17c display the cumulative Brier score *difference* relative to the GBM baseline over the backtest period. The GBM reference is fixed at zero (red dashed line); values below zero indicate that the SYNTH model outperforms GBM cumulatively, with **more negative values denoting a larger advantage**. In all three competitions, both SYNTH models remain well below the GBM reference throughout the evaluation window, confirming that the miner-based ensembles provide materially superior probabilistic forecasts compared to the parametric benchmark—not merely at a few isolated time points, but consistently across the full evaluation period.

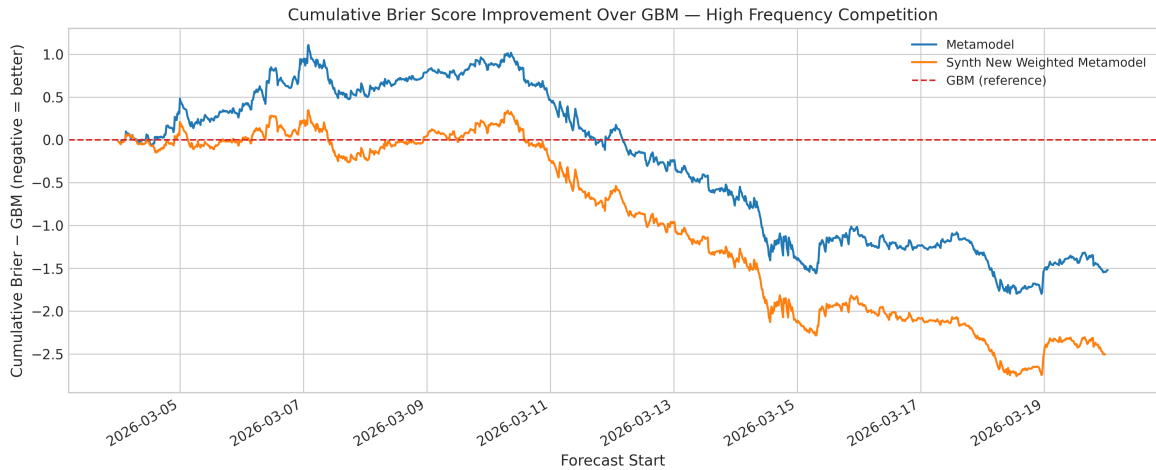
The Synth Weighted Metamodel achieves a more negative cumulative difference than the Synth Classic Metamodel across all competitions, reflecting the consistent upweighting of the strongest-performing miners within each rolling optimisation window.



(a) 24-hour competition. Cumulative Brier difference vs. GBM; more negative = better.



(b) HFT 60-minute competition. Cumulative Brier difference vs. GBM; more negative = better.



(c) HFT 15-minute competition. Cumulative Brier difference vs. GBM; more negative = better.

Figure 17: Cumulative Brier score difference relative to the GBM baseline across all three competitions. The red dashed line at zero represents the GBM reference; more negative values indicate a larger advantage over GBM. Both SYNTH models remain consistently below zero throughout the evaluation window, and the Synth Weighted Metamodel achieves the most negative cumulative difference in all competitions.

5.2.3 Summary Tables

Tables 4, 5, and 6 provide aggregate scoring summaries for all models across the full backtest windows.

The aggregate statistics confirm the picture emerging from the trajectory and cumulative Brier analyses. In the **24-hour competition** (Table 4), both SYNTH models outperform GBM on all reported metrics, with the Synth Weighted Metamodel achieving the best performance on mean Brier score and mean log-likelihood. In the **HFT competitions** (Tables 5 and 6), the improvement over GBM is even more pronounced, and the Synth Weighted Metamodel’s advantage over the Synth Classic Metamodel is also more marked—particularly in the 60-minute market where the larger variance in price trajectories over the longer horizon provides more room for a well-calibrated distributional forecast to add value.

Table 4: Aggregate prediction-market scoring summary — 24-hour competition. $N = 15,441$ observations. Best values in bold.

Model	Total LL	Mean LL	Total Brier	Mean Brier	LL Impr. vs GBM	Brier Impr. vs GBM
GBM Baseline	−5645.279	−0.3656	1786.608	0.1157	+0.00%	+0.00%
Synth Classic Metamodel	−5463.565	−0.3538	1785.313	0.1156	+3.22%	+0.07%
Synth Weighted Metamodel	−5390.340	−0.3491	1730.877	0.1121	+4.52%	+3.12%

Table 5: Aggregate prediction-market scoring summary — HFT 60-minute competition. $N = 22,949$ observations. Best values in bold.

Model	Total LL	Mean LL	Total Brier	Mean Brier	LL Impr. vs GBM	Brier Impr. vs GBM
GBM Baseline	−10753.758	−0.4686	3634.565	0.1584	+0.00%	+0.00%
Synth Classic Metamodel	−10690.862	−0.4659	3591.577	0.1565	+0.58%	+1.18%
Synth Weighted Metamodel	−10592.458	−0.4616	3576.066	0.1558	+1.50%	+1.61%

Table 6: Aggregate prediction-market scoring summary — HFT 15-minute competition. $N = 22,949$ observations. Best values in bold.

Model	Total LL	Mean LL	Total Brier	Mean Brier	LL Impr. vs GBM	Brier Impr. vs GBM
GBM Baseline	−11969.085	−0.5216	4039.130	0.1760	+0.00%	+0.00%
Synth Classic Metamodel	−11805.476	−0.5144	4013.192	0.1749	+1.37%	+0.64%
Synth Weighted Metamodel	−11733.885	−0.5113	3998.961	0.1743	+1.97%	+0.99%

6 Discussion

6.1 Implications for the Competition

Remark 6.1 (Metamodel weights are separate from miner rewards). The Synth Weighted Metamodel affects only how SYNTH’s downstream products aggregate miner paths into a forecast distribution. Miner **rewards** continue to be determined entirely by the existing CRPS-based leaderboard mechanism, exactly as before. The metamodel weights introduced here do *not* feed back into emission allocations.

The introduction of the Synth Weighted Metamodel marks a step-change in how SYNTH aggregates miner outputs. Rather than passively pooling all highly-rewarded paths, the new system actively identifies the miners whose variance forecasts are most informative and concentrates weight on them. This creates a more discriminating selection criterion for metamodel inclusion: miners that provide genuine forecasting diversity—not merely high average CRPS—are specifically valued for the ensemble.

The finding that the optimiser typically selects *one* representative per operator cluster is an important structural result. It means that running multiple near-identical miners provides rapidly diminishing returns under the new weighting regime: operators who currently deploy many clones will find that only one of their miners receives meaningful metamodel weight. This has the potential to reshape operator strategy toward differentiation—a beneficial externality for the network’s overall forecasting quality.

6.2 Implications for Miners

Miners seeking to maximise their contribution to the Synth Weighted Metamodel ensemble should focus on two qualities:

1. **Variance forecast accuracy:** the ability to predict the realized hourly volatility of asset returns, especially during regime transitions and high-volatility episodes.
2. **Forecasting distinctiveness:** submitting predictions that are genuinely different from those of other miners. A model that accurately forecasts volatility *and* does so in a way that is not already captured by other miners will attract disproportionately large weight.

Miners running sophisticated volatility models—such as GARCH-type models (Bollerslev, 1986; Engle, 1982) or machine-learning approaches that condition on recent realized volatility—are likely to receive the highest metamodel weights.

6.3 Implications for Traders

For traders who use SYNTH forecasts as inputs to trading strategies, the Synth Weighted Metamodel offers several advantages over the classic baseline:

1. **Better-calibrated distributions:** the PIT analysis demonstrates that SYNTH’s distributional forecasts are now closer to the true return distribution, reducing the risk of systematic mis-pricing in options or volatility strategies.
2. **More stable probability estimates:** the concentrated, diversity-aware weighting reduces dependence on idiosyncratic miner paths, producing more reliable signals for market-making or arbitrage strategies.
3. **More accurate tail coverage:** the improved tail calibration directly benefits risk managers who use Value-at-Risk or Expected Shortfall estimates derived from SYNTH’s distributions.

4. **Lower noise:** by down-weighting redundant miners, the Synth Weighted Metamodel reduces the idiosyncratic noise that can corrupt ensemble probability estimates, resulting in a cleaner signal.

6.4 Future Directions

This work opens several avenues for further research and product development:

Continuous model improvement. The current optimisation window is fixed at 10 days for the 24-hour competition and 5 days for the HFT competition. Adaptive schemes that dynamically adjust the window length based on market regime (e.g., narrowing during fast-moving markets to capture recent volatility dynamics) are an active area of investigation.

Multi-objective weighting. The current design optimises a single criterion (QLIKE on hourly variance). Future versions may incorporate additional objectives—such as prediction-market calibration metrics or CRPS on longer horizons—into a multi-objective framework, balancing accuracy against coverage across the full forecast horizon.

Ensemble combination methods. The literature on forecast combination (Genre et al., 2013; Timmermann, 2006) suggests that simple equal-weight schemes sometimes outperform optimised combinations out of sample due to estimation error. Regularised variants of the QLIKE optimisation—such as L_2 penalisation of large weight deviations from a uniform prior—may improve out-of-sample stability while preserving the diversity advantage of the new scheme.

Leveraging operator identity. The clustering analysis (Section 2) identifies operator groups with high precision. Explicitly incorporating operator identity into the weighting procedure—for example, by first aggregating within-cluster and then weighting across clusters—could provide a structurally cleaner diversity mechanism.

Richer calibration metrics. The PIT analysis presented here evaluates calibration at fixed horizons. Extending to conditional calibration metrics—assessing whether the forecast distribution remains well-calibrated conditional on market regimes, asset classes, or time of day—would provide a more comprehensive picture and could feed directly into the optimisation objective.

Acknowledgements

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